

Numerical Methods

03.12.2024

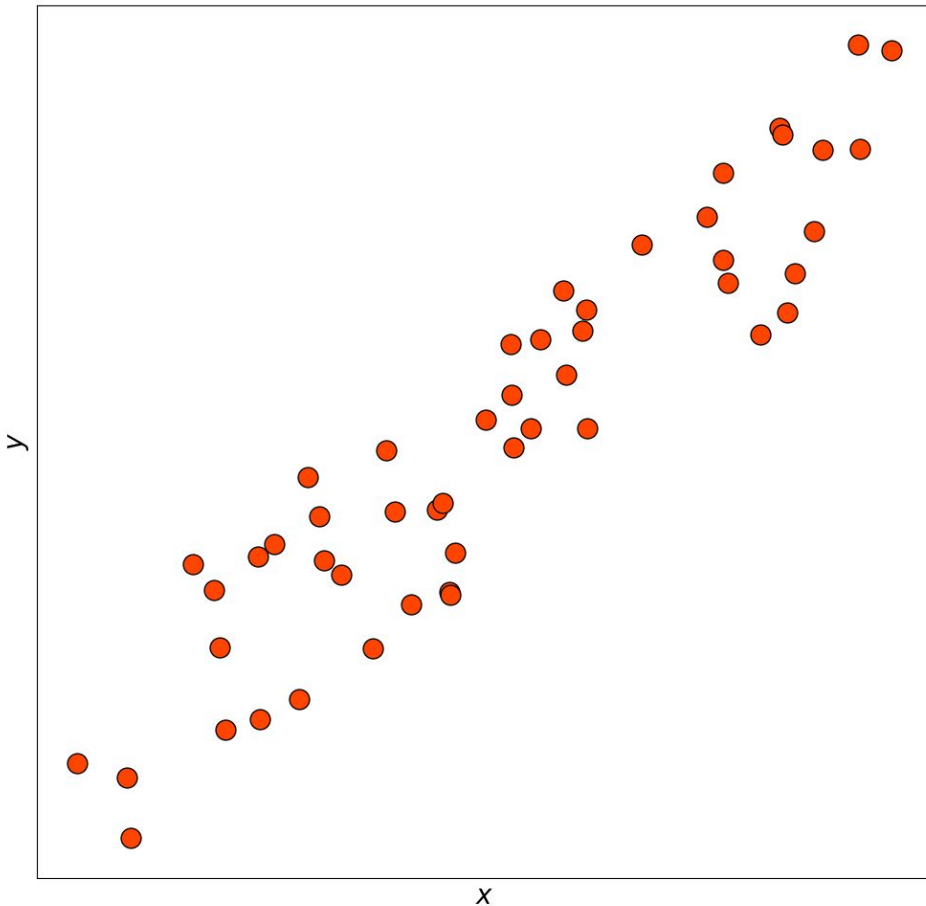
Kristina Kislyakova

Fitting models to observational data



Introduction

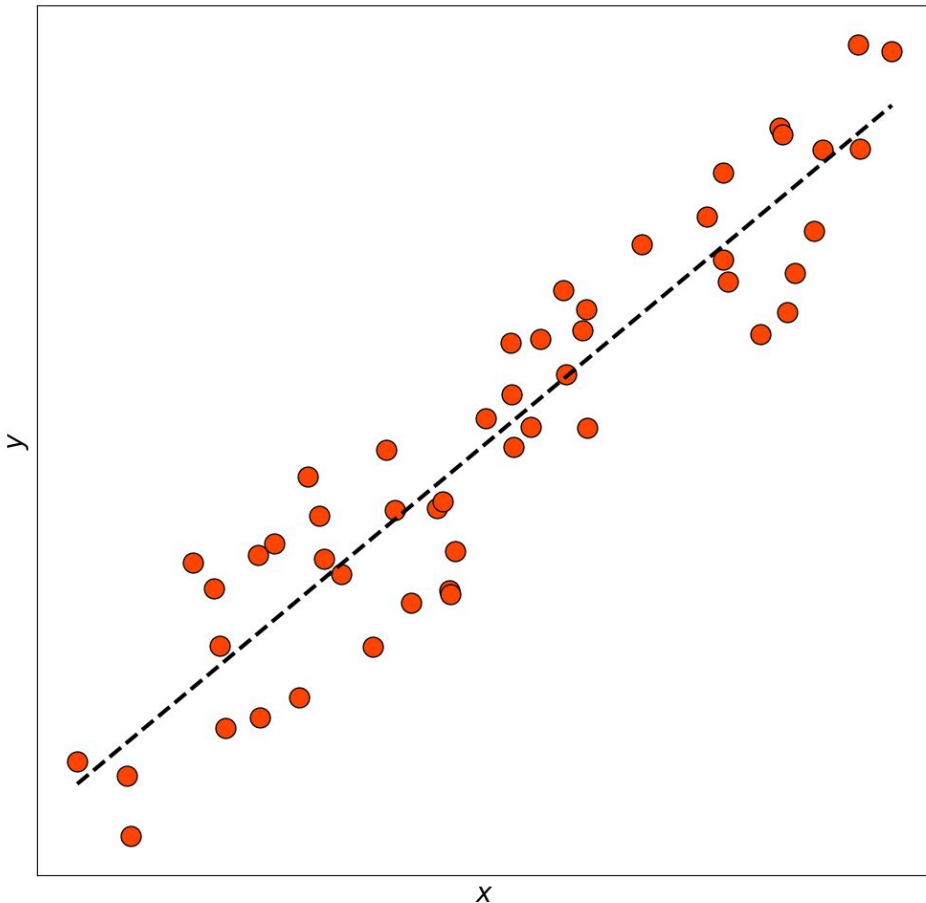
Given a set of observations, we often want to condense the information it contains by fitting it to a “model” that depends on adjustable parameters.



Introduction

Basic approach:

1. choose/design a ***figure-of-merit function*** measuring the agreement between the data and the model with a particular choice of parameters (convention: small values = close agreement).
2. adjust the model parameters to achieve a ***minimum*** in the merit function, i.e., find the best-fit parameters → this is a minimization problem in many dimensions



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Some important issues arise, related to the fact that data are generally not exact (***measurement errors!***).

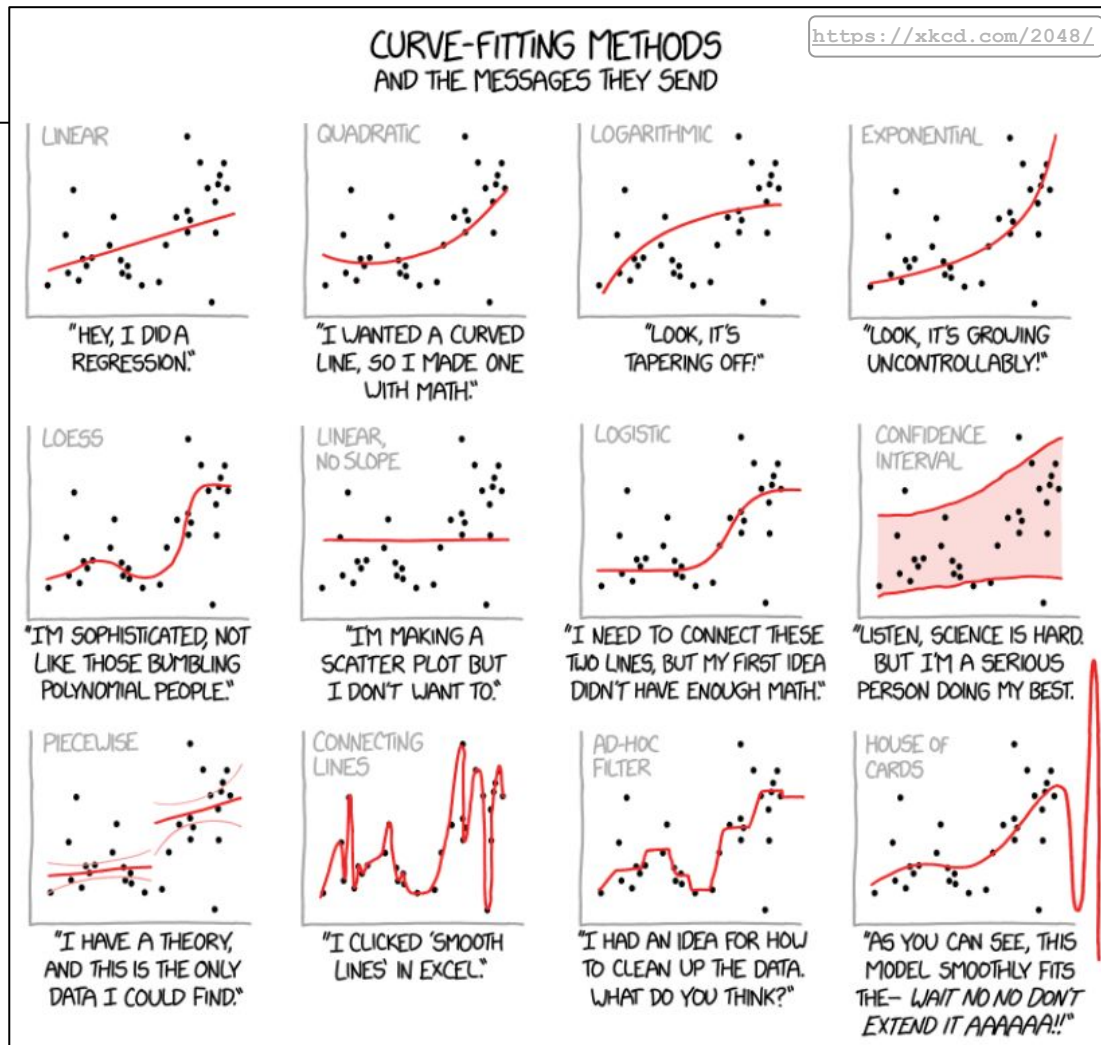
This means that data never exactly fit the model, even when that model is correct. We need to assess if the model is appropriate: we need to test the ***goodness-of-fit***.

We also need to know the accuracy with which parameters are determined by the data set – in practice: we need to know the likely ***errors of the best-fit parameters***.

Introduction

Therefore, to be useful, a fitting procedure should provide the following elements:

1. parameters,
2. error estimates on the parameters,
3. a statistical measure of goodness-of-fit.



Least squares as a maximum likelihood estimator

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$$y(x) = y(x; a_1, \dots, a_M)$$

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The first thing that comes to mind is the least-squares fit:

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Where does this come from?

Maximum likelihood estimators!

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Given a particular data set, we have the intuitive feeling that some parameter sets $a_1 \dots a_M$ are very unlikely while others may be very likely. How can we quantify this? How can we select fitted parameters that are “most likely” to be correct?

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We identify the probability of the data given the parameters as the likelihood of the parameters given the data. At this point, we can then decide to fit for the parameters $a_1 \dots a_M$ by finding the values that maximize the likelihood defined in this way.

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This type of estimation is ***maximum likelihood estimation***.

Least squares as a maximum likelihood estimator

Suppose that each data point y_i has a measurement error that is independently random and distributed as a Gaussian distribution around the “true” model $y(x)$, and that the standard deviations σ of these distributions are the same for all points. The probability of the data set is the product of the probabilities of each point:

$$P \propto \prod_{i=1}^N \left\{ \exp \left[-\frac{1}{2} \left(\frac{y_i - y(x_i)}{\sigma} \right)^2 \right] \Delta y \right\}$$

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Least-squares fitting is a maximum likelihood estimation of the fitted parameters **if** the measurement errors are independent and normally distributed with constant standard deviation.

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For errors that actually are normally distributed, the quantity χ^2 is a sum of N squares of normally distributed quantities, each normalized to unit variance. Once we have adjusted the $a_1 \dots a_M$ to minimize the value of χ^2 , the terms in the sum are not all statistically independent. For models that are linear in the parameters, the probability distribution for different values of χ^2 at its minimum can be derived analytically, and is the ***chi-square distribution for $N - M$ degrees of freedom***.

Chi square probability distribution

$P(\chi^2|\nu)$ is defined as the probability that the observed chi-square for a correct model should be less than a value χ^2 . Its complement $Q(\chi^2|\nu)$ is the probability that the observed chi-square will exceed the value χ^2 by chance even for a correct model. In both cases ν is an integer, the number of degrees of freedom. The functions have the limiting values:

$$P(0|\nu) = 0 \qquad P(\infty|\nu) = 1$$

$$Q(0|\nu) = 1 \qquad Q(\infty|\nu) = 0$$

and the following relation to the incomplete gamma functions:

$$P(\chi^2|\nu) = P\left(\frac{\nu}{2}, \frac{\chi^2}{2}\right) = \frac{\gamma(\nu/2, \chi^2/2)}{\Gamma(\nu/2)}$$

$$Q(\chi^2|\nu) = Q\left(\frac{\nu}{2}, \frac{\chi^2}{2}\right) = 1 - P\left(\frac{\nu}{2}, \frac{\chi^2}{2}\right) = \frac{\Gamma(\nu/2, \chi^2/2)}{\Gamma(\nu/2)}$$

$$\text{scipy.special.gammaincc}(a, x) \longrightarrow P(a, x) = \frac{1}{\Gamma(a)} \int_0^x t^{a-1} e^{-t} dt$$

$$\text{scipy.special.gammainc}(a, x) \longrightarrow Q(a, x) = \frac{1}{\Gamma(a)} \int_x^\infty t^{a-1} e^{-t} dt$$

$$\text{scipy.special.gamma}(z) \longrightarrow \Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt$$

Example

Suppose that we have $\chi^2 = 10.82$ and $v = 9$. We can compute:

```
P = scipy.special.gammaincc(9/2, 10.82/2)
```

```
Q = scipy.special.gammainc(9/2, 10.82/2)
```

And we find that:

```
P = 0.28825
```

```
Q = 0.71175
```

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If Q is a very small probability for a data set, then the apparent discrepancies are unlikely to be chance fluctuations. Much more probably:

1. the model is wrong — can be statistically rejected
2. you are considering a wrong size for the measurement errors σ_i — they are really larger than stated
3. the measurement errors may not be normally distributed — this is a fairly common possibility, and also fairly benign.

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A rule of thumb is that a “typical” value of χ^2 for a “moderately” good fit is $\chi^2 \approx \nu$. More precise is the statement that the χ^2 statistic has a mean ν and a standard deviation $\sqrt{2\nu}$, and, asymptotically for large ν , becomes normally distributed.

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We consider the problem of fitting a set of N data points (x_i, y_i) to a straight-line model

$$y(x) = y(x; a, b) = a + bx$$

This problem is called **linear regression**. We assume that the uncertainty σ_i associated with each measurement y_i is known, and that the values x_i of the dependent variable are known exactly.

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$$\chi^2(a, b) = \sum_{i=1}^N \left(\frac{y_i - a - bx_i}{\sigma_i} \right)^2$$

If the measurement errors are normally distributed, this merit function gives maximum likelihood parameter estimations of a and b .

If the errors are not normally distributed, the estimations are not maximum likelihood, but may still be useful in a practical sense.

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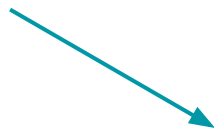
$$S \equiv \sum_{i=1}^N \frac{1}{\sigma_i^2} \quad S_x \equiv \sum_{i=1}^N \frac{x_i}{\sigma_i^2} \quad S_y \equiv \sum_{i=1}^N \frac{y_i}{\sigma_i^2}$$
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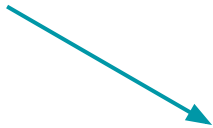
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$$aS_x + bS_{xx} = S_{xy}$$


$$\Delta \equiv SS_{xx} - (S_x)^2$$
$$a = \frac{S_{xx}S_y - S_xS_{xy}}{\Delta}$$
$$b = \frac{SS_{xy} - S_xS_y}{\Delta}$$

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$$\sigma_a^2 = S_{xx} / \Delta$$
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$$r_{ab} = \frac{-S_x}{\sqrt{SS_{xx}}}$$

A positive value of r_{ab} indicates that the errors in a and b are likely to have the same sign, while a negative value indicates that the errors are anticorrelated, likely to have opposite signs.

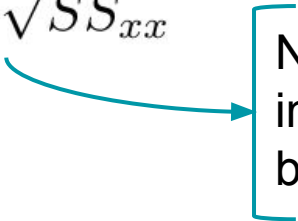
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Note that this is the correlation between the uncertainty in a and the uncertainty in b , it is **not** the correlation between the considered sample of x_i and y_i !!!

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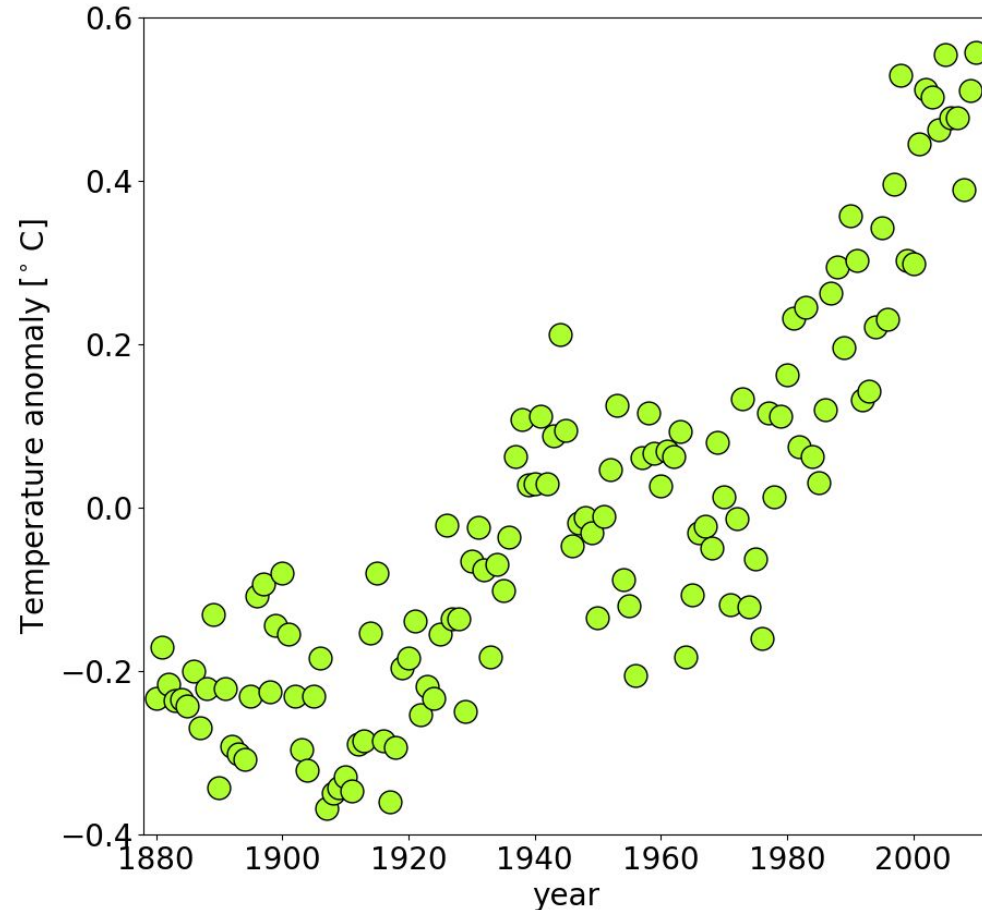
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- If Q is larger than ~ 0.1 , then the goodness-of-fit is believable.
- If it is larger than ~ 0.001 , then the fit may be acceptable if the errors are non-normal or have been moderately underestimated.
- If $Q < 0.001$ then the model and/or the estimation procedure can (should!) be called into question.

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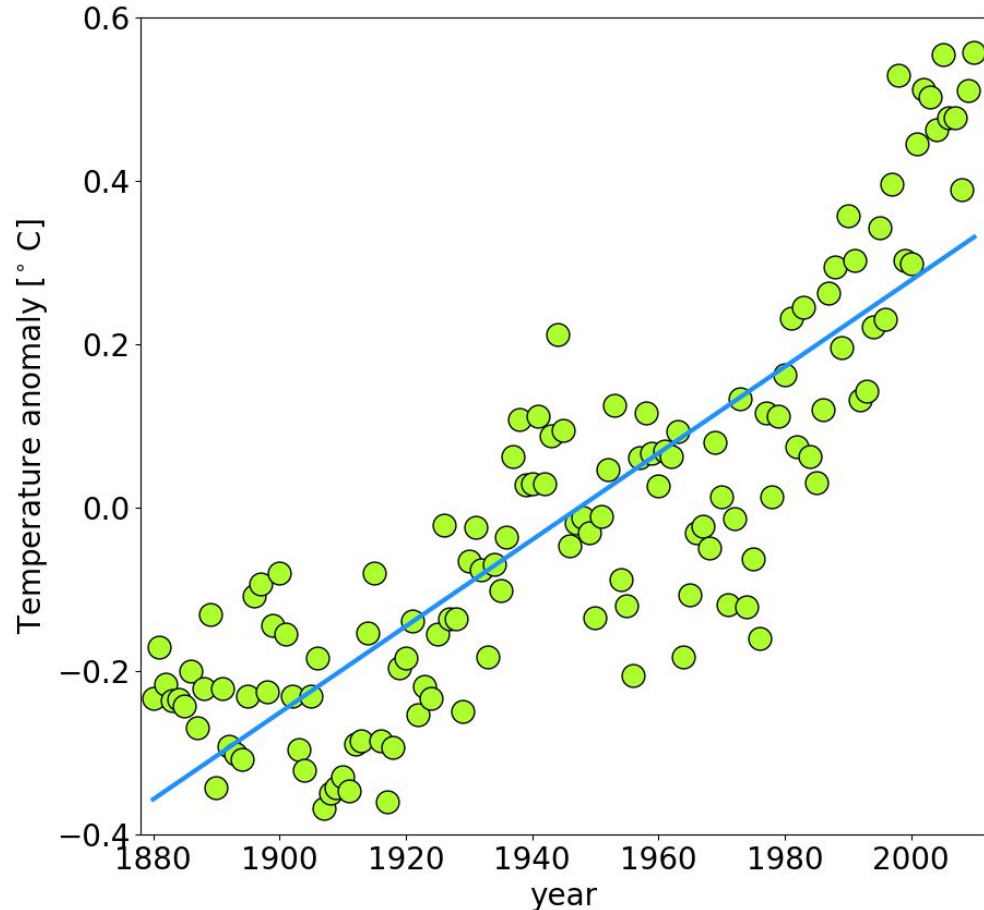
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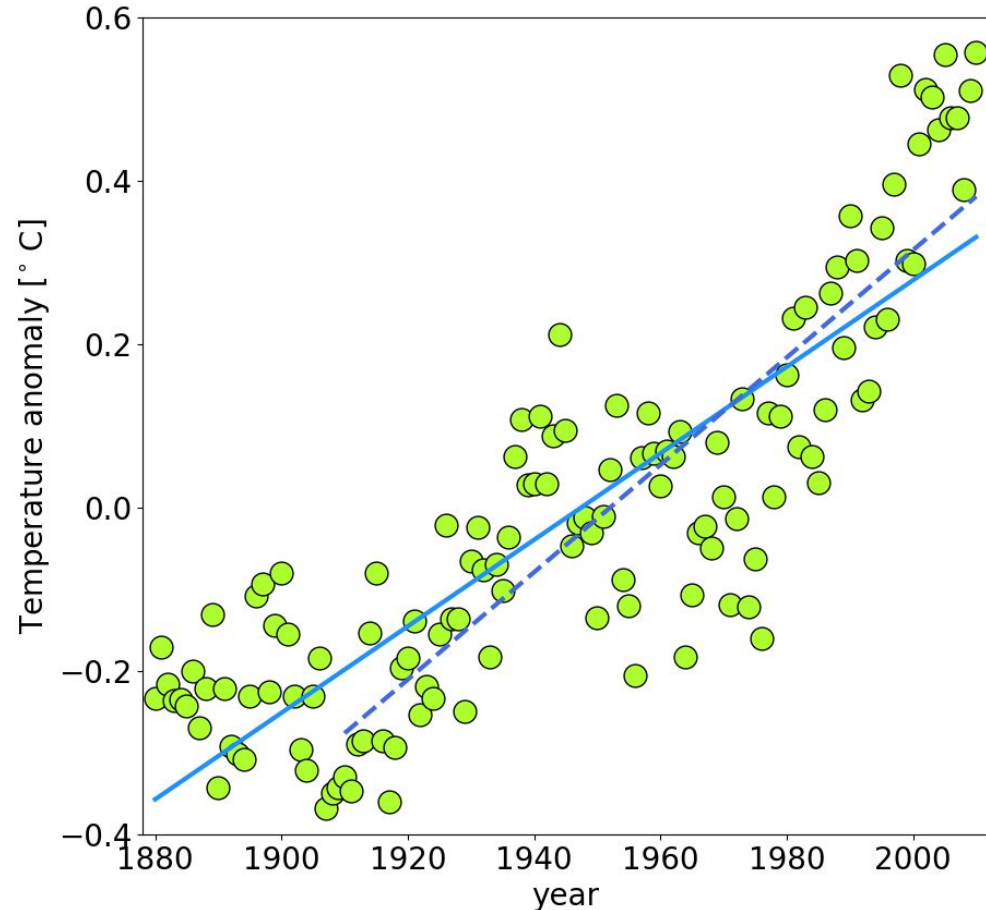
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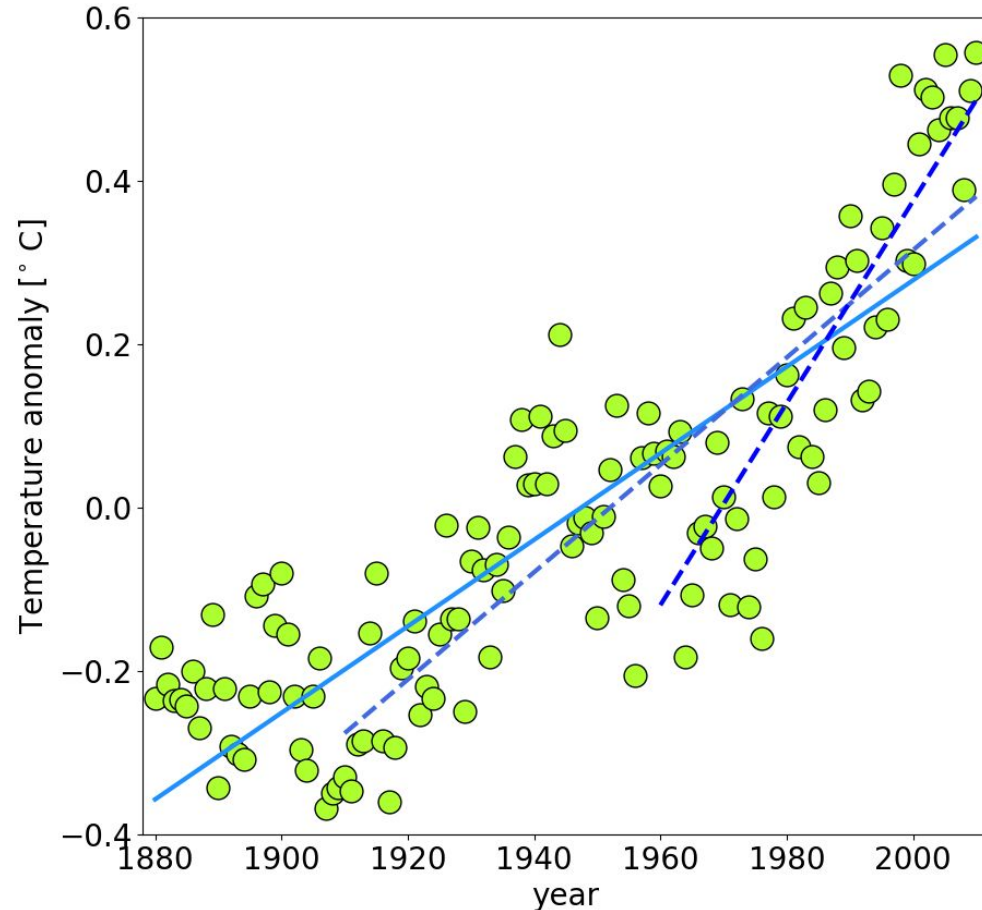
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$0.12^{\circ}/\text{decade}$ (1960-2010)



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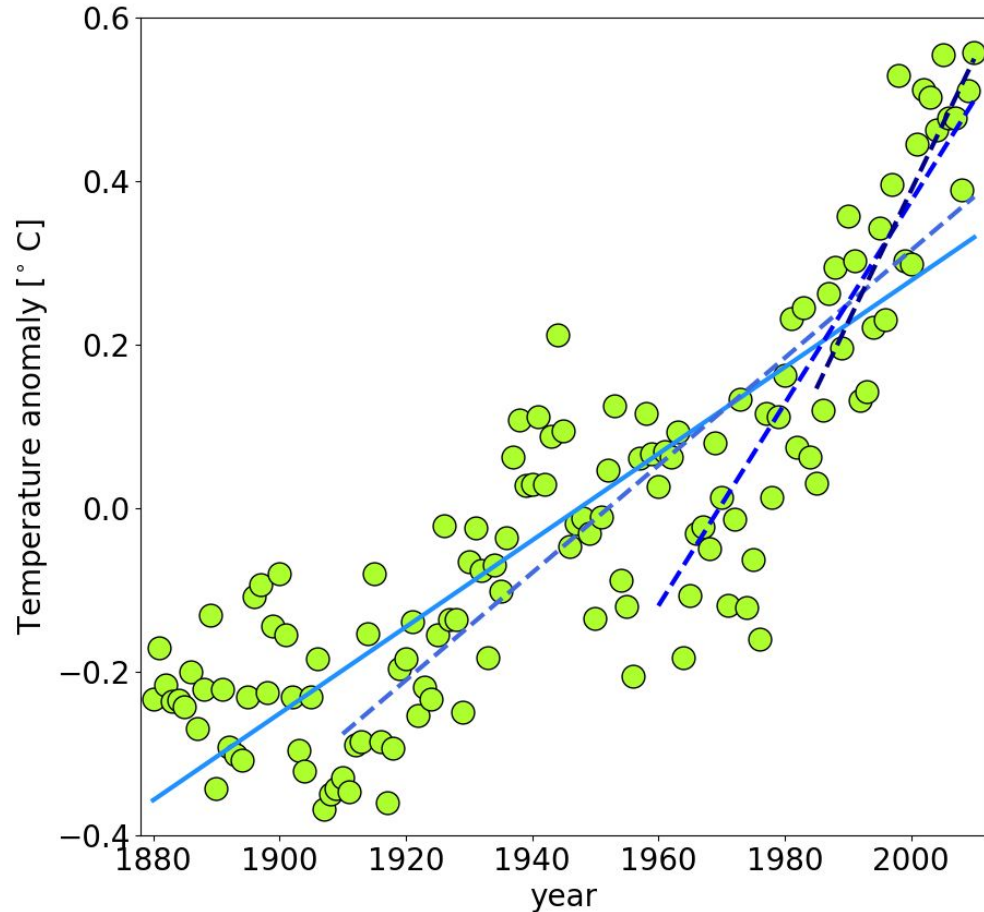
Rate of increase:

$0.05^{\circ}/\text{decade}$ (1880-2010)

$0.07^{\circ}/\text{decade}$ (1910-2010)

$0.12^{\circ}/\text{decade}$ (1960-2010)

$0.16^{\circ}/\text{decade}$ (1985-2010)

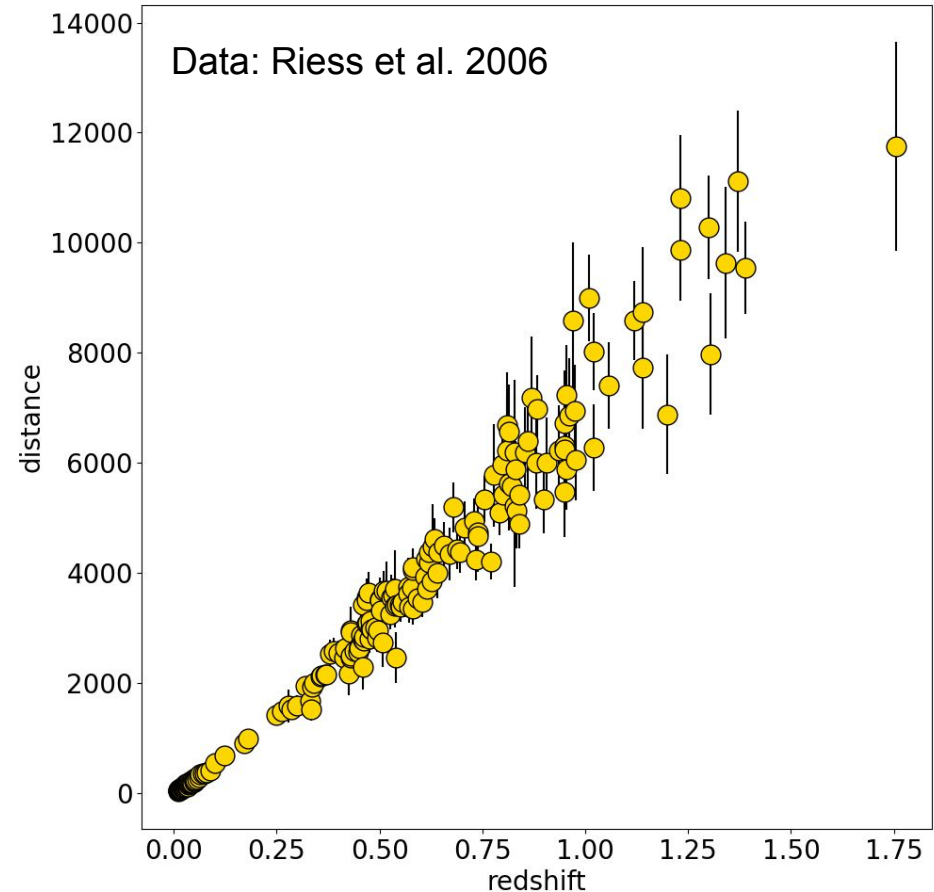


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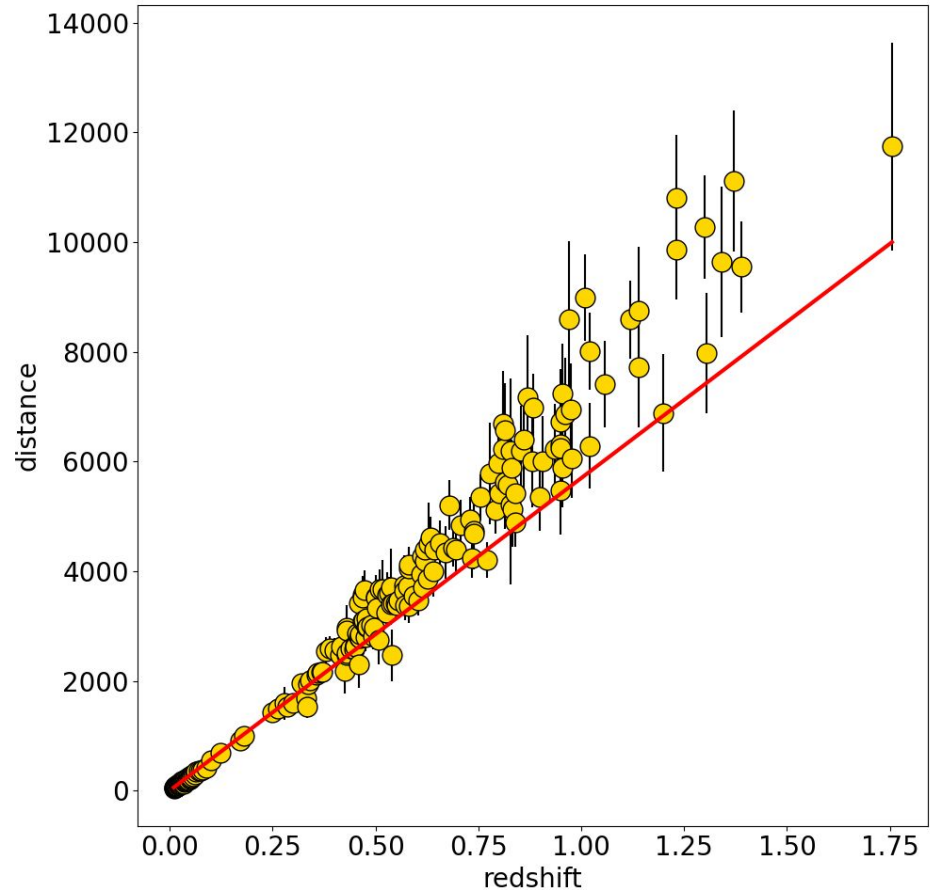


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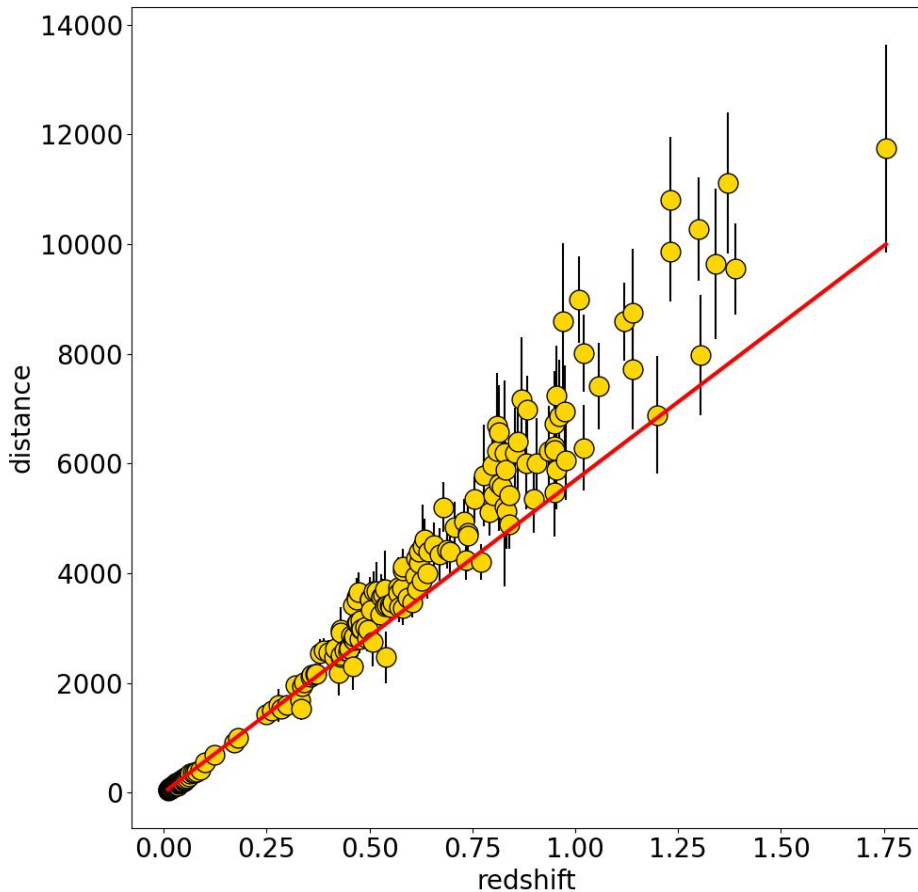


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BUT: $Q \sim 0!$

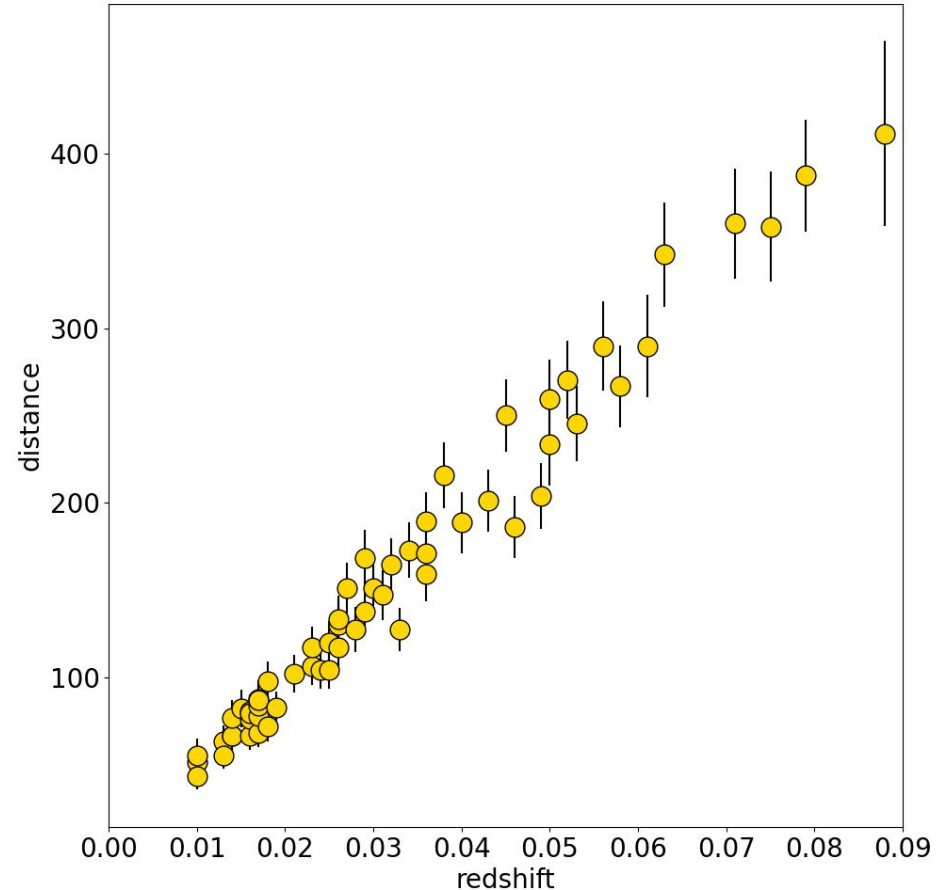


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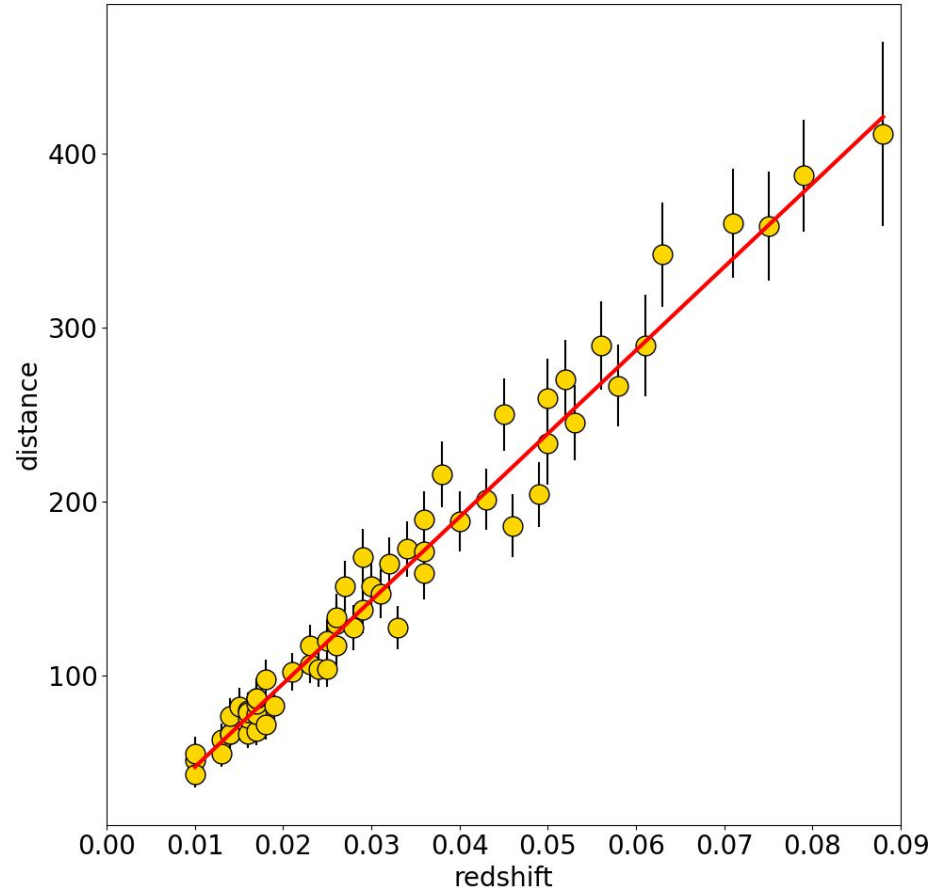
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$$H_0 = 62.7 \text{ (km/s)/Mpc} \text{ [Q = 0.78]}$$



Straight line data with errors in both coordinates

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$$\chi^2(a, b) = \sum_{i=1}^N \frac{(y_i - a - bx_i)^2}{\sigma_{y_i}^2 + b^2 \sigma_{x_i}^2}$$

where σ_{x_i} and σ_{y_i} are, respectively, the x and y standard deviations for the i -th point.

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where σ_{x_i} and σ_{y_i} are, respectively, the x and y standard deviations for the i -th point. The weighted sum of variances in the denominator can be understood both as the variance in the direction of the smallest χ^2 between each data point and the line with slope b , and also as the variance of the linear combination $y_i - a - bx_i$ of two random variables x_i and y_i ,

$$\text{Var}(y_i - a - bx_i) = \text{Var}(y_i) + b^2 \text{Var}(x_i) = \sigma_{y_i}^2 + b^2 \sigma_{x_i}^2 \equiv 1/w_i$$

The sum of the square of N random variables, each normalized by its variance, is thus χ^2 -distributed.

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Therefore, a reasonable way to proceed is to consider a general one-dimensional function to minimize with respect to b , while using this expression at each stage to ensure that the minimum with respect to b is also minimized with respect to a .

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Standard errors

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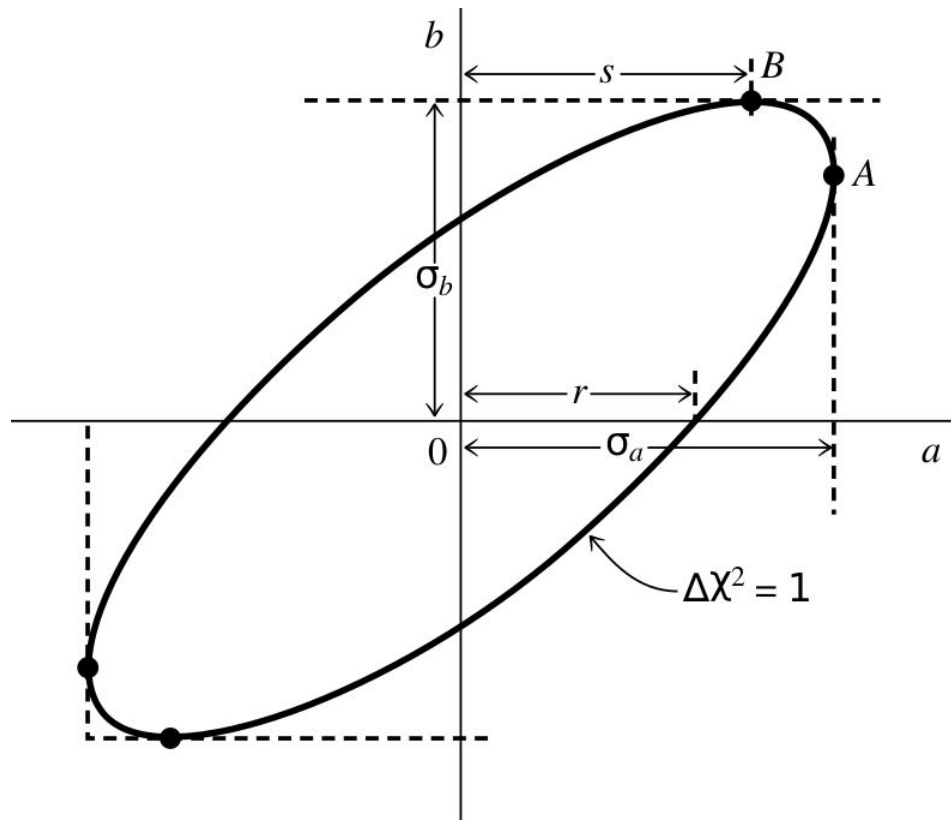
Standard errors

Finding the standard errors σ_a and σ_b on the parameters a and b is a bit more complicated. Later we will see that, in appropriate circumstances, the standard errors in a and b are the projections onto the a and b axes of the “confidence region boundary” where χ^2 takes on a value larger than its minimum by a quantity $\Delta\chi^2 = 1$.

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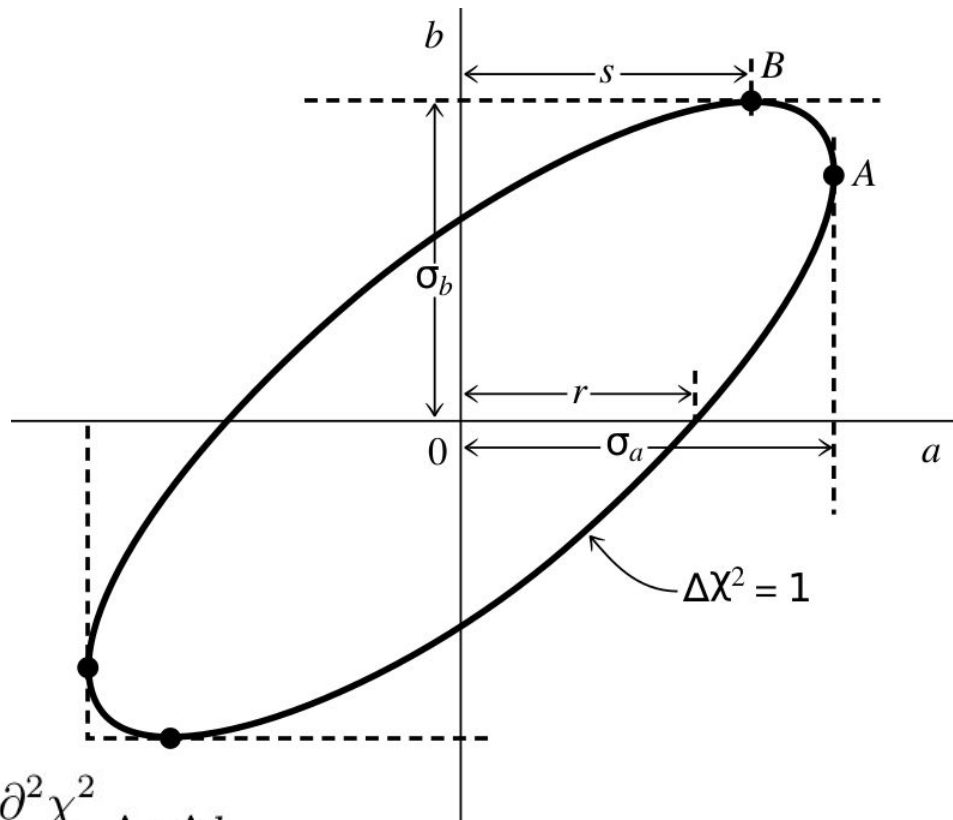


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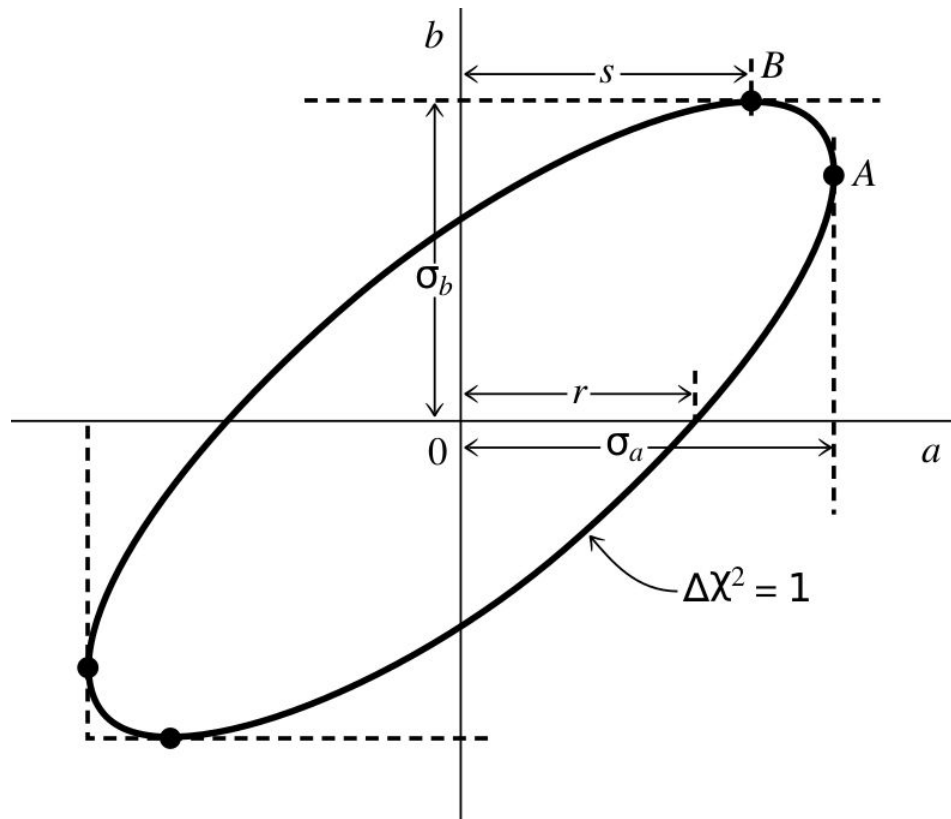
$$\Delta\chi^2 \approx \frac{1}{2} \left[\frac{\partial^2 \chi^2}{\partial a^2} (\Delta a)^2 + \frac{\partial^2 \chi^2}{\partial b^2} (\Delta b)^2 \right] + \frac{\partial^2 \chi^2}{\partial a \partial b} \Delta a \Delta b$$



Straight line data with errors in both coordinates

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In the case we are considering now, because of the non-linearity in b , analytic formulas for the second derivatives are quite unwieldy and, more importantly, the lowest-order term frequently gives a poor approximation to $\Delta\chi^2$, so we cannot proceed in this way.

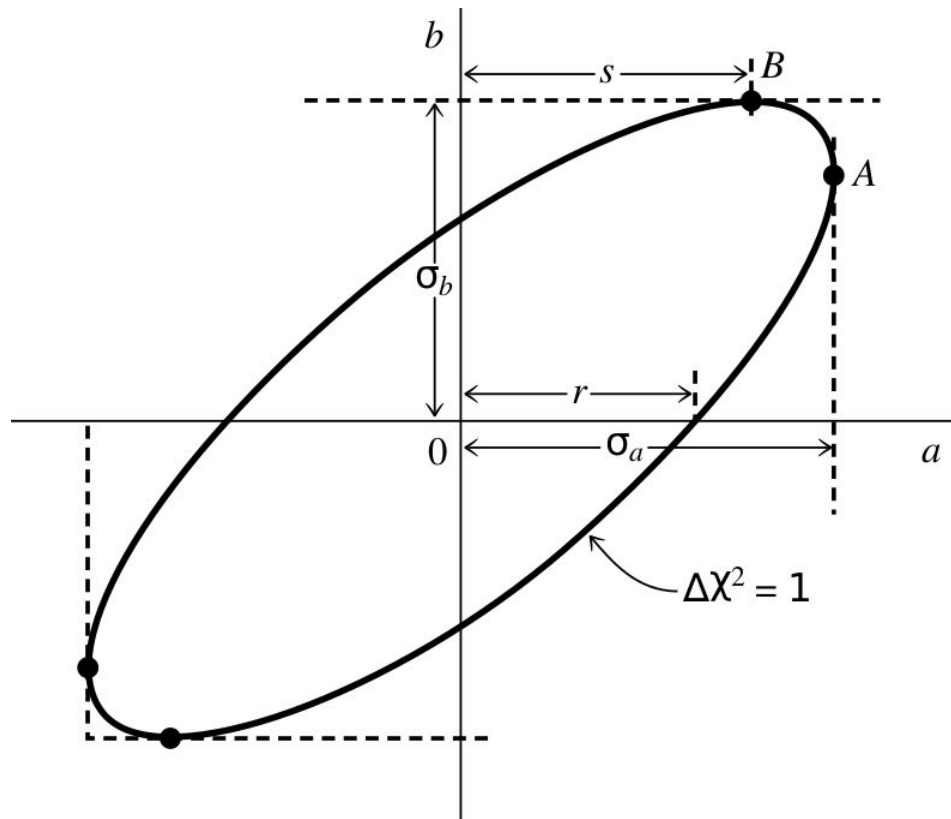


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Therefore, we need to find the roots of $\Delta\chi^2 = 1$ numerically, by adjusting the value of the slope b away from the minimum.



General linear least squares

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$$y(x) = a_1 + a_2x + a_3x^2 + \dots + a_Mx^{M-1}$$

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The general form of this kind of model is:

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As before, σ_i is the measurement error of the i -th data point. Once again, we will pick as best parameters those that minimize χ^2 .

General linear least squares

Confidence intervals

General linear least squares

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Let's consider that for our data the value of χ^2 is a minimum at $\mathbf{a}_0$. If parameter values are perturbed away from this point, χ^2 increases.

The region within which χ^2 increases by no more than a set amount $\Delta\chi^2$ defines some M -dimensional confidence region around $\mathbf{a}_0$. There are choices of $\Delta\chi^2$ for which the region contains 68%, 90%, ... of probability distribution for the parameters.

These regions are the confidence regions for the parameters $\mathbf{a}_0$.

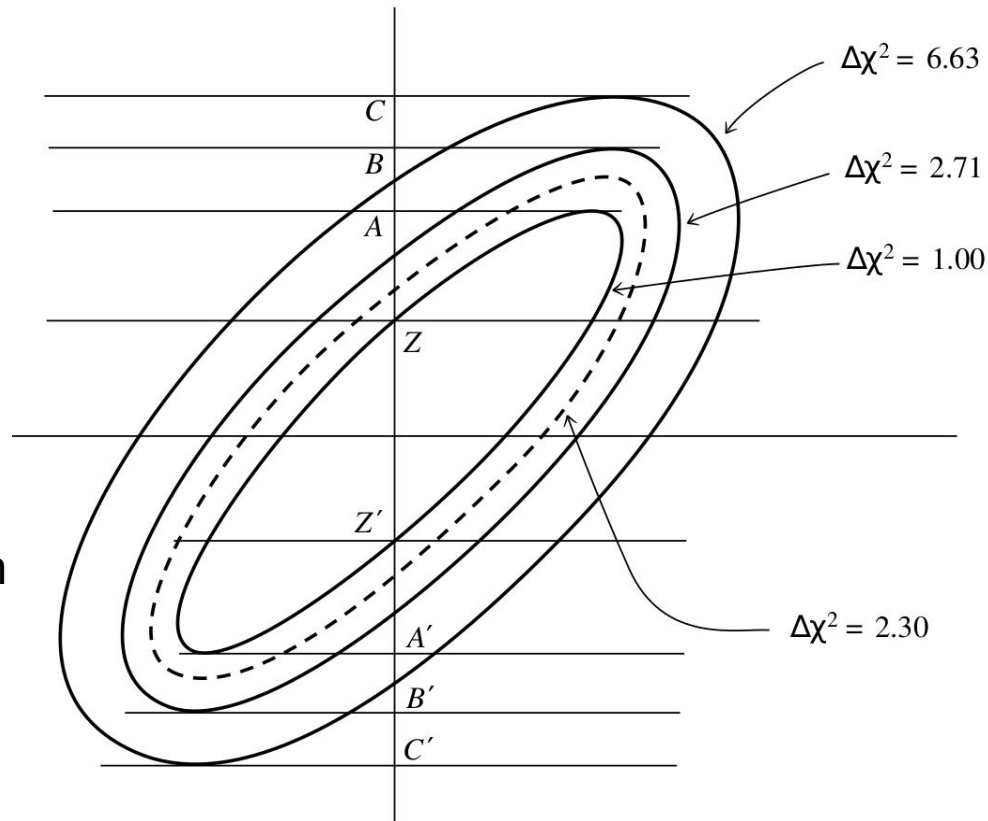
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Python built-in functions

numpy.polyfit

`numpy.polyfit(x, y, deg, rcond=None, full=False, w=None, cov=False)` [\[source\]](#)

Least squares polynomial fit.

Note

This forms part of the old polynomial API. Since version 1.4, the new polynomial API defined in `numpy.polynomial` is preferred. A summary of the differences can be found in the [transition guide](#).

Fit a polynomial $p(x) = p[0] * x^{deg} + \dots + p[deg]$ of degree *deg* to points (*x*, *y*). Returns a vector of coefficients *p* that minimises the squared error in the order *deg*, *deg*-1, ... 0.

The `Polynomial.fit` class method is recommended for new code as it is more stable numerically. See the documentation of the method for more information.

Python built-in functions

scipy.stats.linregress

`scipy.stats.linregress(x, y=None, alternative='two-sided')`

[\[source\]](#)

Calculate a linear least-squares regression for two sets of measurements.

Parameters: `x, y` : *array_like*

Two sets of measurements. Both arrays should have the same length. If only `x` is given (and `y=None`), then it must be a two-dimensional array where one dimension has length 2. The two sets of measurements are then found by splitting the array along the length-2 dimension. In the case where `y=None` and `x` is a 2x2 array, `linregress(x)` is equivalent to `linregress(x[0], x[1])`.

alternative : {'two-sided', 'less', 'greater'}, *optional*

Defines the alternative hypothesis. Default is 'two-sided'. The following options are available:

- 'two-sided': the slope of the regression line is nonzero
- 'less': the slope of the regression line is less than zero
- 'greater': the slope of the regression line is greater than zero

New in version 1.7.0.

Python built-in functions

scipy.optimize.curve_fit

```
scipy.optimize.curve_fit(f, xdata, ydata, p0=None, sigma=None,  
absolute_sigma=False, check_finite=True, bounds=(- inf, inf), method=None,  
jac=None, **kwargs)
```

[\[source\]](#)

Use non-linear least squares to fit a function, *f*, to data.

Assumes $ydata = f(xdata, *params) + \epsilon$.

Parameters: *f* : *callable*

The model function, $f(x, \dots)$. It must take the independent variable as the first argument and the parameters to fit as separate remaining arguments.

xdata : *array_like or object*

The independent variable where the data is measured. Should usually be an M-length sequence or an (k,M)-shaped array for functions with k predictors, but can actually be any object.

ydata : *array_like*

The dependent data, a length M array - nominally $f(xdata, \dots)$.